

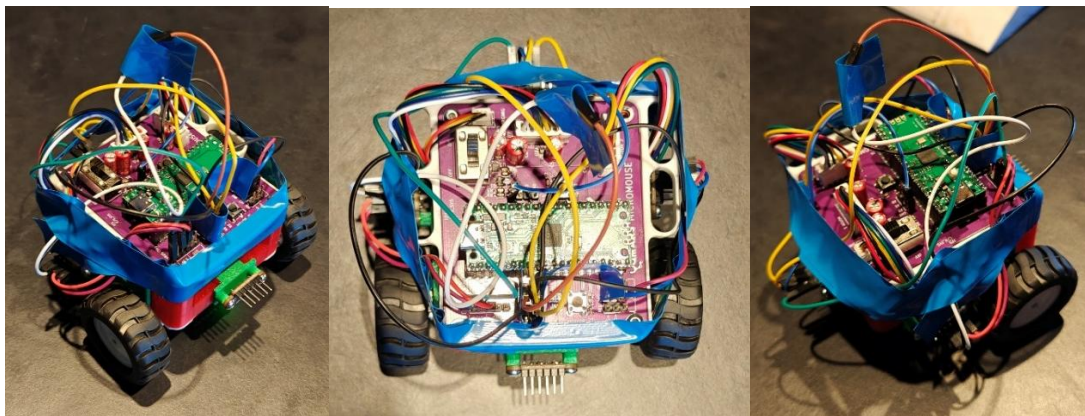
Micromouse - 2026

Team Members:

- Genevieve White
- Niklas Dark
- Baryal Mansoor
- William Davis

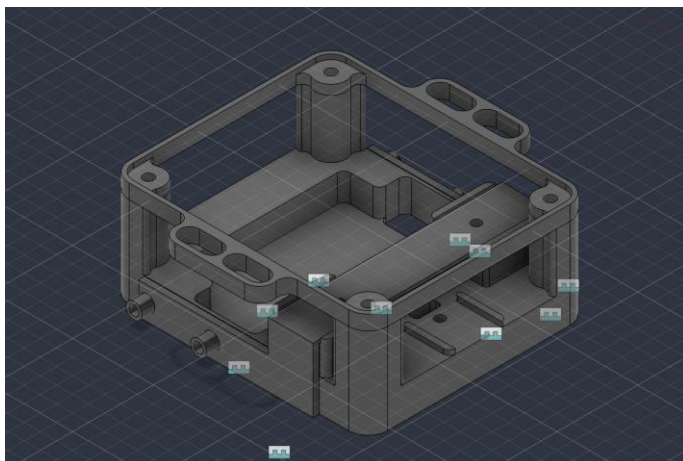
Images:

Below are images of the final micromouse design. A large part of the chassis and design of the mouse was unchanged from the micromouse kit used however, slight changes were made. The original infrared sensors were switched out with time of flight sensors which calculated the distance an object was away from it opposed to giving a Boolean readout of whether an object was within a certain distance. To accommodate these new sensors slight modifications to the chassis were made so they could be mounted correctly.

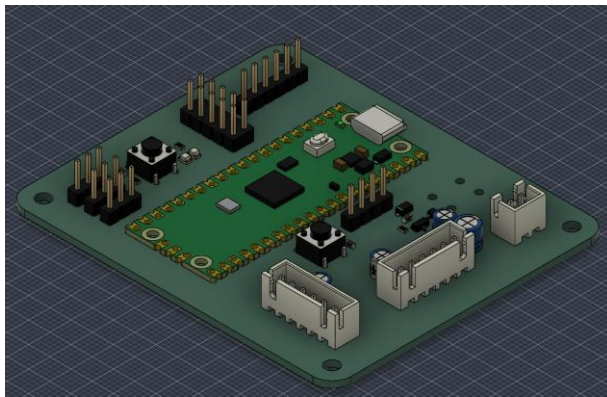


Cad designs:

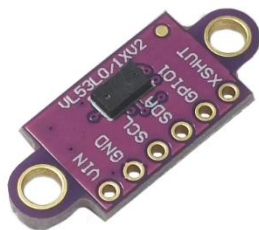
Chassis



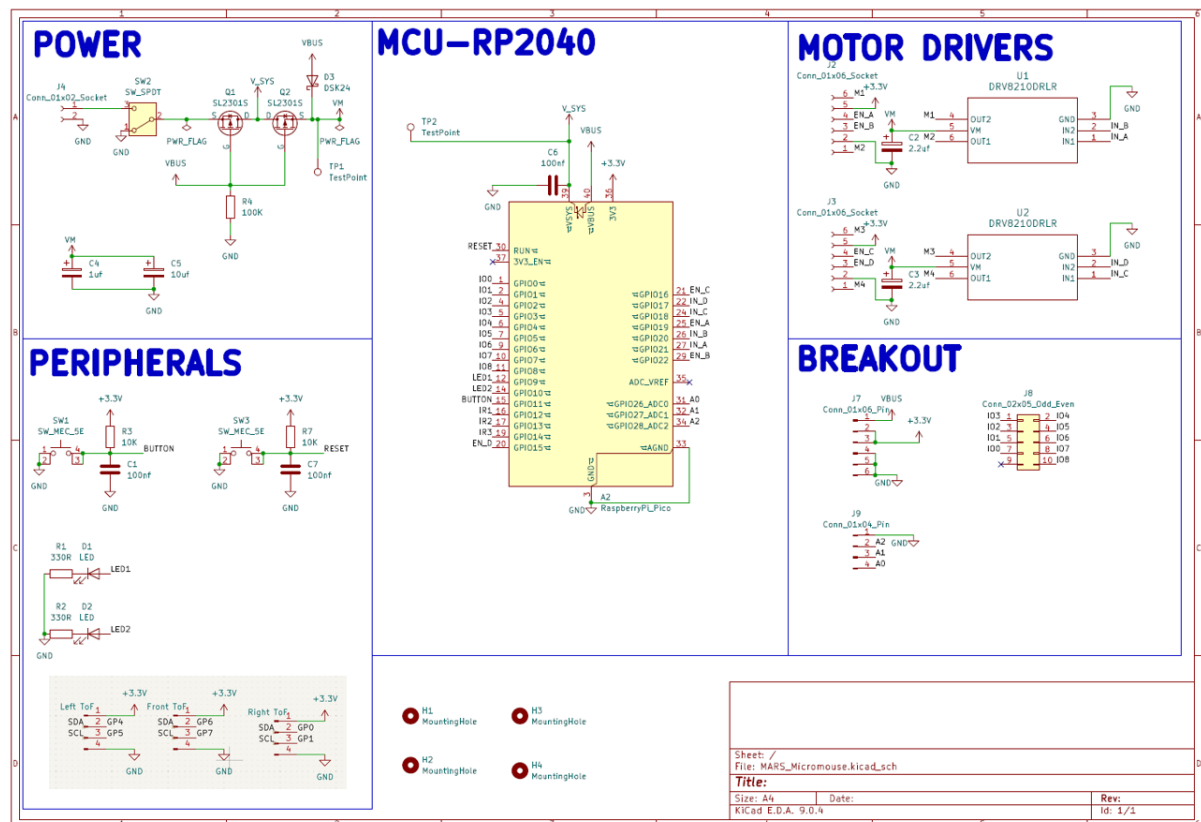
PCB



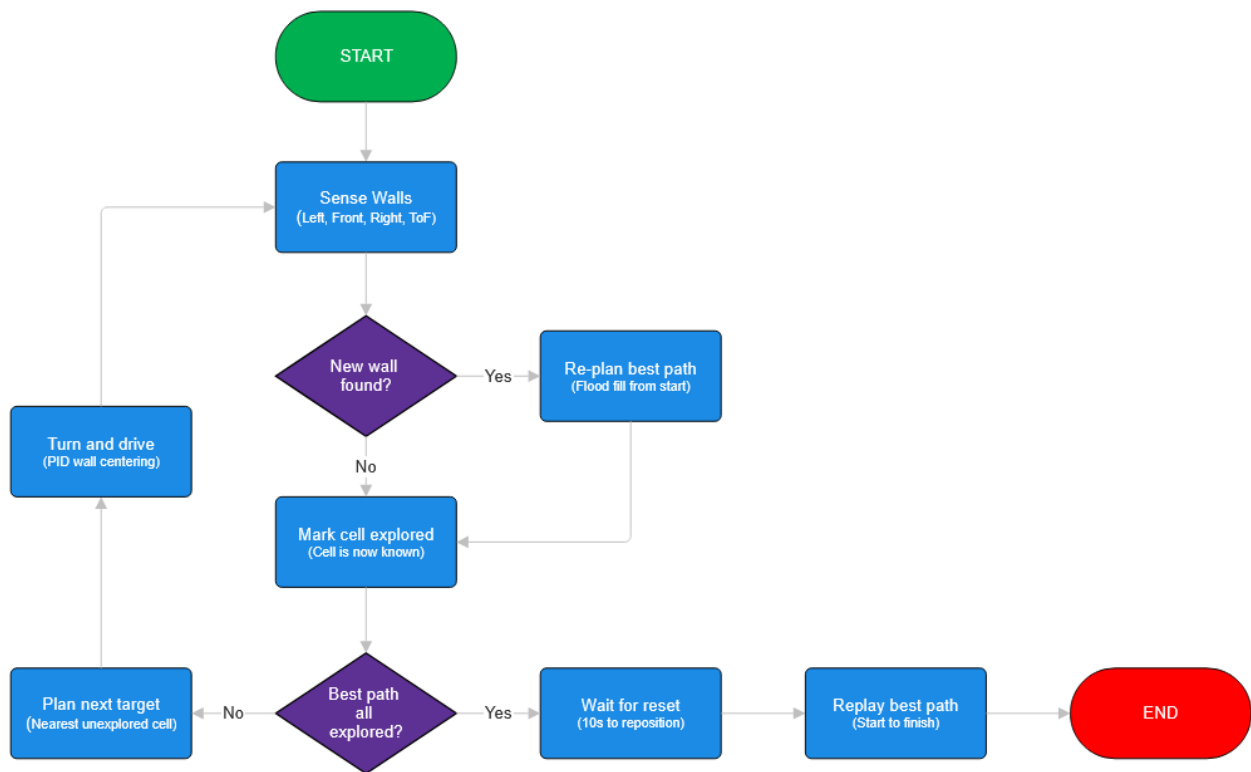
Time of flight sensor



Circuit Schematics:



Algorithm flowcharts:



Cost:

Item	Quantity	Cost per unit	Total
Micromouse Kit	1	\$40.00	\$40.00
Time of flight sensors	4	\$4.11	\$16.45
Total Cost			\$56.45

4Pcs Time-Of-Flight ToF Ranging Sensor Breakout 940Nm -VL53L0XV2 Distance Module IIC

Brand: Utopially

\$16⁴⁵

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- 1.The from is a time-of-flight ranging system integrated into a compact module. This board is a carrier for the , so we recommend careful reading of the datasheet (1MB pdf) before using this product.
- 3.Ranging measurements are available through the sensor's I²C (TWI) interface, which is also used to configure sensor settings, and the sensor provides two additional pins: a shutdown input and an interrupt output.
- 2.The VL53L0 uses 's FlightSense technology to precisely measure how long it takes for emitted pulses of infrared light to reach the nearest object and be reflected back to a detector, so it can be considered a tiny, self-contained lidar system.
- 4.The is a great IC, but its small, leadless, LGA package makes it difficult for the typical student or hobbyist to use. It also operates at a recommended voltage of 2.8 V, which can make interfacing difficult for microcontrollers operating at 3.3 V or 5 V. Our breakout board addresses these issues, making it easier to get started using the sensor, while keeping the overall size as small as possible.
- 5.A time-of-flight ranging system integrated into a compact module

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